

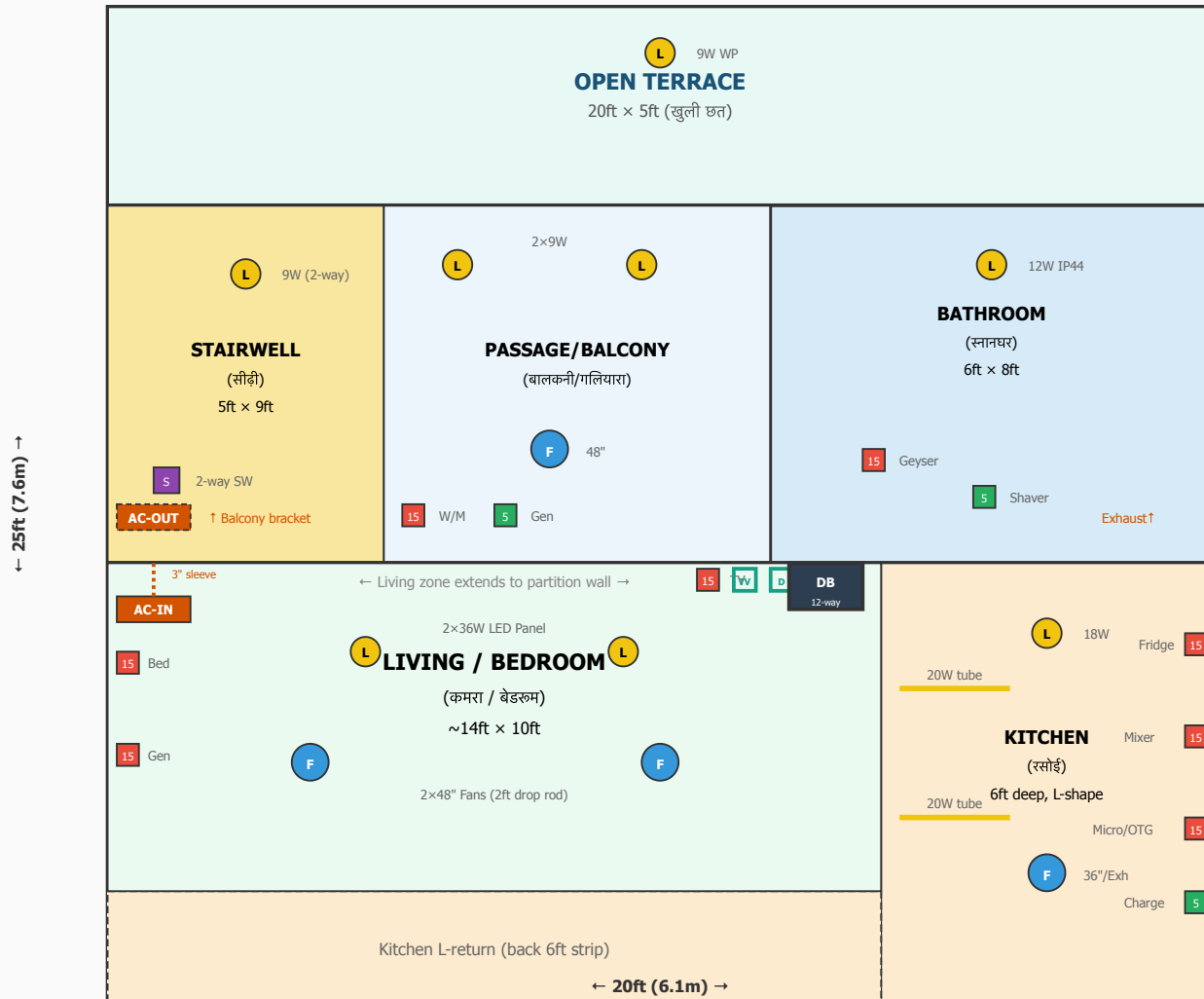
पहली मंजिल — Electrical Design

First Floor Electrical Layout & Specifications | Building: 20ft × 25ft | Floor Height: 12ft

1. Electrical Layout Plan (विद्युत लेआउट)

* Light Point ⚡ Ceiling Fan ■ 15A Socket ○ 5A Socket □ Switch Board AC AC Unit DB Distribution Board TV TV/Data Point

Scale: 1 grid = 1ft approx



2. Lighting & Fan Schedule (लाइटिंग एवं पंखा अनुसूची)

2.1 Lighting Schedule

S.No.	Room / Area	Type	Wattage	Qty	Mounting	Special Notes
1	Living/Bedroom (कमरा)	LED Panel (Slim)	36W	2	False ceiling - recessed	Concealed in gypsum FC; dimmable optional
2	Living/Bedroom	LED Strip (cove)	~15W/m	1 run	False ceiling cove	Warm white 3000K, peripheral cove
3	Kitchen (रसोई)	LED Tube (T5)	20W	2	Under wall-cabinet	Counter illumination; in PVC FC
4	Kitchen	LED Ceiling Light	18W	1	PVC false ceiling	Center of kitchen
5	Bathroom (स्नानघर)	LED Panel (IP44)	12W	1	PVC false ceiling	Waterproof, IP44 rated mandatory
6	Balcony/Passage (बालकनी)	LED Bulb (B22)	9W	2	Ceiling batten holder	Weatherproof if exposed
Total Lighting Points:				10	Total Load: ~220W (lighting only)	

S.No.	Room / Area	Type	Wattage	Qty	Mounting	Special Notes
7	Staircase Landing (सीढ़ी)	LED Bulb (B22)	9W	1	Wall bracket / ceiling	2-way switch (GF + 1F); conduit pre-embedded
8	Open Terrace (छत)	LED Bulb (weatherproof)	9W	1	Weatherproof fixture	IP65 rated; timer/sensor optional
Total Lighting Points:				10	Total Load: ~220W (lighting only)	

2.2 Fan Schedule (पंखा)

S.No.	Room / Area	Type	Size	Wattage	Qty	Mounting	Notes
1	Living/Bedroom	Ceiling Fan	48" (1200mm)	75W	2	2ft drop rod from slab (through FC)	BLDC preferred (35W); hook in RCC slab
2	Kitchen	Ceiling Fan / Exhaust	36" (900mm)	50W	1	Ceiling / wall exhaust	Exhaust fan (12") if ceiling height issue
3	Balcony	Ceiling Fan	48" (1200mm)	75W	1	Direct mount or short rod	Outdoor-rated if exposed to elements
Total Fans:				4	Total Load: ~275W (fans)		

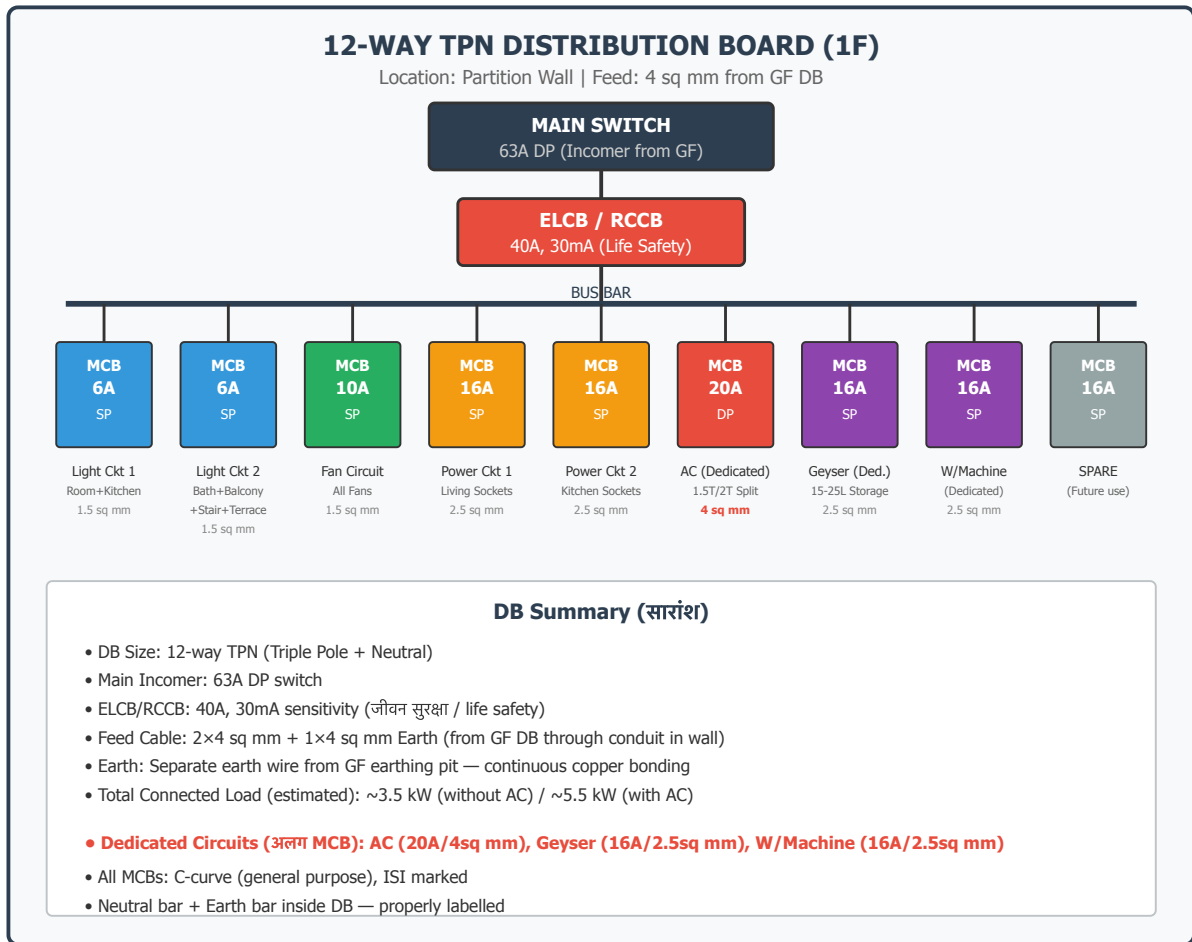
3. Power Socket Schedule (सॉकेट अनुसूची)

S.No.	Room	Position / Wall	Rating	Purpose (उपयोग)	Height from FFL	Circuit	Wire Size
1	Living/Bedroom	Partition wall (TV unit area)	15A + 5A combo	TV + Set-top box + WiFi router	4ft (behind TV)	Power Ckt 1	2.5 sq mm
2	Living/Bedroom	Left wall (bed side)	15A + 5A combo	Phone charging, lamp, general	1ft (bed height)	Power Ckt 1	2.5 sq mm
3	Living/Bedroom	Left wall (middle)	15A	General (iron, vacuum, etc.)	1.5ft	Power Ckt 1	2.5 sq mm
4	Kitchen	Right wall (bottom)	15A	Fridge (फ्रिज) - dedicated	1ft	Power Ckt 2	2.5 sq mm
5	Kitchen	Right wall (counter height)	15A	Mixer/Grinder (मिक्सर)	3.5ft	Power Ckt 2	2.5 sq mm
6	Kitchen	Right wall (counter height)	15A	Microwave / OTG	3.5ft	Power Ckt 2	2.5 sq mm
7	Kitchen	Back wall	5A	Phone charging, small appliance	3.5ft	Power Ckt 2	1.5 sq mm
8	Bathroom	Near shower (high)	15A	Geyser (गीजर) - DEDICATED	6ft	Geyser (Ded.)	2.5 sq mm
9	Bathroom	Near mirror/basin	5A	Shaver, hair dryer	4ft	Power Ckt 1	1.5 sq mm
10	Balcony	Near bathroom wall	15A	Washing Machine (वॉशिंग मशीन) - DEDICATED	1.5ft	W/M (Ded.)	2.5 sq mm
11	Balcony	Passage wall	5A	General (charging, lamp)	3ft	Power Ckt 1	1.5 sq mm
Total Socket Points:			11 points (7 nos. 15A + 4 nos. 5A)				

Data/TV/WiFi Points: TV coaxial + RJ45 data point at TV unit location (partition wall, 4ft height). WiFi router position near DB for central coverage. Total: 2 data points + 1 TV point.

4. Distribution Board — Single Line Diagram (DB विवरण)

DB Location: Inside room, on partition wall (between living and kitchen), 5ft from floor. Away from moisture zones. Fed from GF DB via 4 sq mm cable in pre-embedded conduit.



5. AC Provision Detail (एसी प्रावधान)

Indoor Unit (इनडोर यूनिट)

- Position:** Left wall of living/bedroom, 7-8ft from floor
- Capacity:** 1.5T or 2T split AC (inverter type recommended)
- Power:** Dedicated 20A MCB, 4 sq mm wire (L+N+E)
- Socket:** 16A dedicated socket behind indoor unit OR hardwired
- Drain:** Condensate drain pipe to nearest outlet (bathroom/balcony)

Outdoor Unit (आउटडोर यूनिट)

- Position:** Balcony wall bracket (left side) OR on open terrace
- Bracket:** MS powder-coated wall bracket, 2ft from floor
- Clearance:** Min 12" from wall for airflow
- Foundation:** Anti-vibration pads on bracket

Pipe Sleeve & Connections

- Wall sleeve:** 3" (75mm) PVC pipe through partition wall
- Slope:** Slight outward slope for condensate drainage
- Contents through sleeve:**
 - Refrigerant copper pipes (liquid + gas)
 - Inter-connecting cable
 - Condensate drain pipe
- Sealing:** Putty/foam seal after installation
- Height:** Same as indoor unit (7-8ft from floor)

Electrical for AC

- Wire: 4 sq mm FR copper (L+N+E) in 25mm conduit
- MCB: 20A DP (Double Pole) C-curve
- Separate from other circuits in DB
- Stabilizer: External, if voltage fluctuation area

6. False Ceiling Layout (फॉल्स सीलिंग)

Area	Material	FC Height (from floor)	Gap from RCC Slab	Lighting in FC	Services in Gap	Notes
Living/Bedroom	Gypsum/POP	9.5 - 10ft	2 - 2.5ft	2×36W recessed LED panels + LED strip in cove	AC piping, wiring, downlighter boxes	L-shape or peripheral design; cove for indirect light
Kitchen	PVC panels (moisture resistant)	9.5ft	2.5ft	1×18W surface LED + under-cabinet tubes separate	Wiring, exhaust duct	Easy maintenance; oil/moisture resistant
Bathroom	PVC panels (waterproof)	8.5 - 9ft	3 - 3.5ft	1×12W IP44 LED panel	Geyser pipes, exhaust, wiring	IP44 minimum; access panel for geyser valve
Balcony/Passage	None (open slab)	—	—	Surface mount / batten holder	—	Conduit on slab surface, painted

False Ceiling Construction Sequence:
1. Mark FC level on all walls (laser level) → 2. Install GI channel frame (periphery + grid) → 3. Pull wiring through conduits to fixture positions → 4. Fix downlighter/panel cutout boxes → 5. Install FC boards → 6. Cut openings for lights/AC vents → 7. Finish (putty + paint for gypsum; clip-in for PVC) → 8. Install fixtures.

Fan hooks must go into RCC slab directly (not into false ceiling frame). Drop rods pass through FC. Fan canopy covers the FC hole.

7. Wiring Specification Table (वायरिंग विवरण)

Circuit	MCB Rating	Wire Size (sq mm)	Wire Type	Conduit Size	No. of Wires	Max Load	Color Code
Main Feed (GF → 1F)	63A (main sw.)	4 sq mm	FR PVC Copper	25mm UPVC	3 (L+N+E)	~5.5 kW	Red/Black/Green
Lighting Circuit 1 (Room+Kitchen)	6A SP	1.5 sq mm	FR PVC Copper	20mm UPVC	3 (L+N+E)	800W	Red/Black/Green
Lighting Circuit 2 (Bath+Balcony+Stair+Terrace)	6A SP	1.5 sq mm	FR PVC Copper	20mm UPVC	3 (L+N+E)	800W	Red/Black/Green
Fan Circuit (All fans)	10A SP	1.5 sq mm	FR PVC Copper	20mm UPVC	3 (L+N+E)	~350W	Red/Black/Green
Power Circuit 1 (Living sockets)	16A SP	2.5 sq mm	FR PVC Copper	20mm UPVC	3 (L+N+E)	3000W	Red/Black/Green
Power Circuit 2 (Kitchen sockets)	16A SP	2.5 sq mm	FR PVC Copper	20mm UPVC	3 (L+N+E)	3000W	Red/Black/Green
AC (Dedicated)	20A DP	4 sq mm	FR PVC Copper	25mm UPVC	3 (L+N+E)	~2000W	Red/Black/Green
Geyser (Dedicated)	16A SP	2.5 sq mm	FR PVC Copper	20mm UPVC	3 (L+N+E)	2000W	Red/Black/Green
Washing Machine (Dedicated)	16A SP	2.5 sq mm	FR PVC Copper	20mm UPVC	3 (L+N+E)	2000W	Red/Black/Green
Stair Light (2-way)	Part of Light Ckt 2	1.5 sq mm	FR PVC Copper	20mm (pre-embedded)	4 (L+N+E+traveller)	—	Red/Black/Green/Yellow

Wire Brand (recommended): Havells Lifeline, Polycab Greenwire, Finolex FR — ISI marked, FR (Flame Retardant) grade mandatory.
Conduit: UPVC rigid conduit (concealed in walls/slab) + flexible corrugated conduit (in false ceiling runs).

8. Material List & Quantities (सामग्री सूची)

8.1 Wires (तार)

Item	Size	Qty (approx)
FR Copper Wire	4 sq mm	30m (2 coils)
FR Copper Wire	2.5 sq mm	90m (1 bundle)
FR Copper Wire	1.5 sq mm	120m (1.5 bundles)
Earth Wire (Green)	2.5 sq mm	60m
Earth Wire (Green)	4 sq mm	15m

8.2 Conduit & Accessories

Item	Size	Qty
UPVC Rigid Conduit	25mm	20m
UPVC Rigid Conduit	20mm	80m
Flexible Conduit (FC runs)	20mm	30m
PVC Junction Boxes	—	15 nos.
Conduit Bends	20mm/25mm	30 nos.
Conduit Clips/Saddles	—	100 nos.
PVC Casing (surface runs)	—	10m

8.4 Switches & Sockets (Modular)

Item	Qty
6-module switch plate (6M)	3
8-module switch plate (8M)	2
4-module switch plate (4M)	4
2-module switch plate (2M)	2
16A socket module	7
6A/5A socket module	4
6A switch module (for lights)	10
6A switch module (for fans)	4
Fan regulator module (electronic)	4
2-way switch module (stair)	2
Bell push switch	1
Indicator module (red)	3
Blank plate / cover	3
TV coaxial socket	1
RJ45 data socket (Cat6)	2

AC Pipe Sleeve	3" (75mm)	1 no. (1ft length)
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8.3 Distribution Board

Item	Specification	Qty
TPN DB Box	12-way, metal/ABS	1
Main Switch	63A DP	1
ELCB/RCCB	40A, 30mA	1
MCB 6A SP	C-curve	2
MCB 10A SP	C-curve	1
MCB 16A SP	C-curve	5
MCB 20A DP	C-curve	1
Bus Bar + Neutral Bar	—	1 set

8.5 Lighting Fixtures

Item	Spec	Qty
LED Slim Panel	36W, recessed, round/square	2
LED Strip + Driver	~15W/m, warm white	1 set (~5m)
LED Tube T5	20W, 4ft	2
LED Ceiling Light	18W, surface/recessed	1
LED Panel IP44	12W, waterproof	1
LED Bulb 9W B22	Cool white	4
Weatherproof fixture (IP65)	For terrace bulb	1
Batten holder (B22)	—	3

8.6 Fans

Item	Spec	Qty
Ceiling Fan 48"	75W / BLDC 35W	3
Ceiling Fan 36" or Exhaust 12"	50W	1
Drop Rod 2ft	For room fans (through FC)	2
Drop Rod 1ft	For balcony fan	1
Fan Hook (ceiling)	Heavy duty, RCC slab	4
Fan Canopy (FC cover)	—	2

8.7 Miscellaneous

Item	Qty
Insulation tape (PVC, multiple colors)	6 rolls
Cable ties (assorted)	1 pack
GI wire (for pulling cables)	50m
Wall plugs + screws	1 box
Geyser (15-25L storage)	1
AC wall bracket (outdoor unit)	1 set
Door bell (wireless/wired)	1

9. Contractor Notes — Work Sequence (ठेकेदार निर्देश — कार्य क्रम)

9.1 Construction Sequence (कार्य अनुक्रम)

Step	Activity	Details (विवरण)	Check Before Next Step
1	Conduit in Walls (दीवारों में नाली)	Chase walls for conduit runs. Embed 20mm/25mm UPVC conduits. Pull GI draw wire. Plug/plaster chases. Conduit from DB location to all switch/socket points.	Pressure test conduits (blow air). Ensure no kinks/blocks. Photo-document before plastering.
2	Conduit in Slab (छत में नाली)	Before slab pour: lay conduit for light/fan points on slab formwork. Install fan hook boxes. Install junction boxes at fixture locations.	Verify all conduit connections. Fan hooks properly embedded. No conduit crushed.
3	AC Sleeve (एसी स्लीव)	Install 3" PVC sleeve through left wall (partition/external) at 7-8ft height. Slight outward slope (5 degrees).	Verify sleeve is clear, not blocked by plaster. Correct position for indoor/outdoor unit.
4	Wall Plastering (प्लास्टर)	Complete wall plastering. Conduit embedded and covered. Switch/socket boxes flush with plaster surface.	All boxes level and flush. No conduit exposed.
5	False Ceiling Frame (फॉल्स सीलिंग फ्रेम)	Install GI channel framework for gypsum FC in living/bedroom. Install PVC FC frame in kitchen/bathroom. Mark and cut openings for fan drop rods.	Level check. Frame strong enough for fixtures. Fan rod holes aligned with slab hooks.
6	Wire Pulling (तार खींचना)	Pull all cables through conduits using GI wire. Connect at junction boxes. Leave adequate loop at fixtures/switches (8-10" extra). Label all wires at DB end.	Continuity test (multimeter) every circuit. Insulation resistance test. No wire damage.
7	FC Board Installation (बोर्ड लगाना)	Fix gypsum/PVC boards on frame. Cut openings for LED panels, downlighters, AC vents, fan canopies. Finish joints (putty for gypsum).	Openings correct size. No wires pinched. Boards secured.
8	DB Installation (DB लगाना)	Mount DB box on partition wall. Connect all circuit wires to respective MCBs. Connect neutral bar, earth bar. Label every MCB clearly.	Tighten all terminal screws (torque). Verify phase-neutral correct. Earth continuity.
9	Fixture Installation (फिक्सचर लगाना)	Install all LED panels, tubes, bulbs, fans, switches, sockets. Connect and test each point individually with DB live.	Every point works. Fans spin correct direction. No sparking. Socket polarity check.
10	Testing & Commissioning (टेस्टिंग)	Full load test. ELCB trip test (press test button). Insulation resistance measurement. Earth loop impedance. Verify 2-way stair switch from both positions.	ELCB trips within 30ms at 30mA. All circuits within safe limits. Issue test certificate.

9.2 Important Safety Notes (सुरक्षा निर्देश)

- **All wire ISI marked + FR grade** (IS 694) — no compromise
 - **Earth wire mandatory** on every circuit — green sleeve
 - **No joints inside conduit** — joints only in junction boxes with proper connectors
 - **Maximum 3 wires in 20mm conduit**, 4 wires in 25mm conduit
 - **Bathroom:** No socket below 4ft height. IP44 minimum for all fittings. Zone rules as per IS 732
 - **Geyser:** Must have dedicated MCB + indicator. Metal body earthed.
- **AC circuit:** 4 sq mm wire mandatory. DP MCB (both L+N break)
 - **Fan hooks:** Direct in RCC slab — NOT in false ceiling frame
 - **Switch height:** 4.5ft from floor (standard). Socket: 1-1.5ft (unless counter-height specified)
 - **DB access:** Must remain accessible at all times. No furniture blocking.
 - **Labelling:** Label every MCB with circuit name (Hindi + English)
 - **Spare conduits:** Leave 1-2 spare conduit runs for future (CCTV, smart home, etc.)

9.3 Coordination Points (समन्वय बिंदु)

Plumber (प्लंबर): Coordinate geyser water inlet/outlet position with electrical point. Washing machine drain + water point adjacent to electrical socket. AC condensate drain routing.

Mason (राजमिस्त्री): Wall chasing for conduits before plaster. Slab conduit before pour. AC sleeve during wall construction.

Carpenter/FC (बढ़ई/FC): FC frame before wire pulling. Fan hole positions. LED panel cutout sizes. Cove dimensions for LED strips.

Painter (पेंटर): Wall painting after switch/socket plate installation (use painter's tape on plates). FC painting before fixture install.

9.4 2-Way Stair Light Wiring (सीढ़ी 2-वे वायरिंग)

Component	Location	Wire	Notes
Switch 1	GF stair base (already installed during GF work)	1.5 sq mm (4 wires: L, N, E, Traveller)	Conduit already in stair wall
Switch 2	1F landing (top of stairs)	Connected via traveller wire to Switch 1	Both switches control same light

Light	Stairwell ceiling (between flights)	Fed from nearest switch	9W LED, wall bracket or ceiling mount
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Document: First Floor Electrical Design | **Building:** 20ft × 25ft Farmhouse | **Prepared for:** Contractor Reference
Standards: IS 732 (Wiring), IS 694 (Cables), IS 3854 (Switches), IS 1293 (MCBs) | **All work to comply with local PWD/electricity board norms**
Print: A4, Portrait | Total estimated load: ~5.5 kW (with AC) | Recommended main feed: 6 kVA